Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program: BSCpE**
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed: 10/22/2026**
Section: CPE11S4	**Date Submitted: 10/22/2026**
Name: CAMBEL, MATTHEW MOAN S.	**Instructor: Engr. Lusung**

1. Objective/s:

This activity aims to navigate and manage files and directories using command line interface.

2. Intended Learning Outcome/s:

The students should be able to:

2.1 Demonstrate how to navigate home and system directories
2.2 Demonstrate how to how to list files and directories.
2.3 Demonstrate how to use globbing to manipulate files and directories.
2.4 Demonstrate how to create, move and delete files and directories.

3. Discussion:

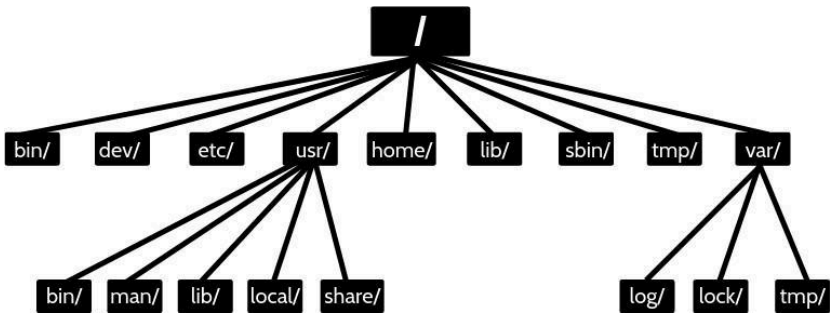
Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.

Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer*.** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory

On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory there is a directory for each user on the system. When a user opens a shell, they should

automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always starts at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

The single period . character always represents the current directory.
Two period .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

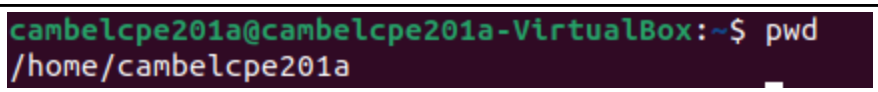
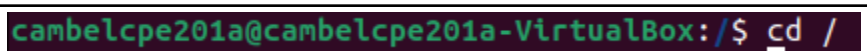
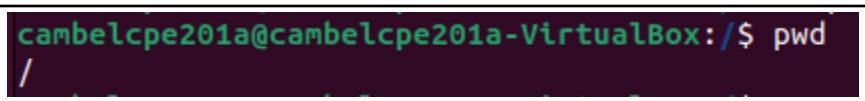
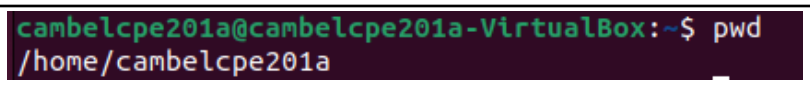
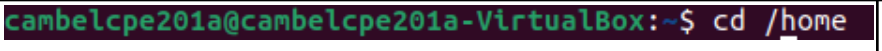
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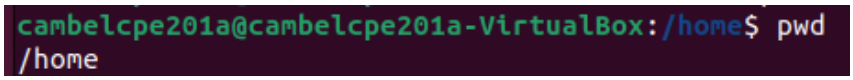
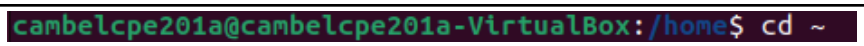
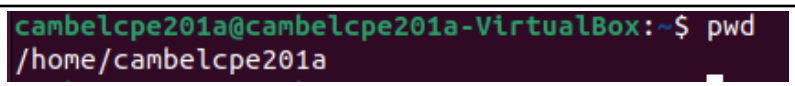
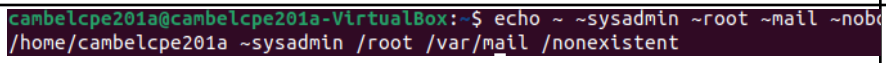
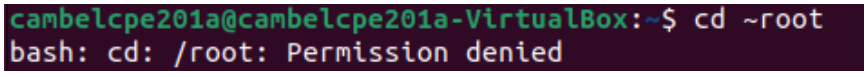
Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

5. Procedure:

1. Login using your username and password.
2. Use terminal emulator application (if you are using desktop version)
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

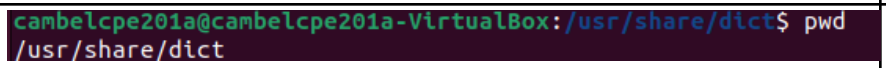
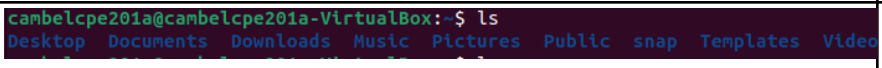
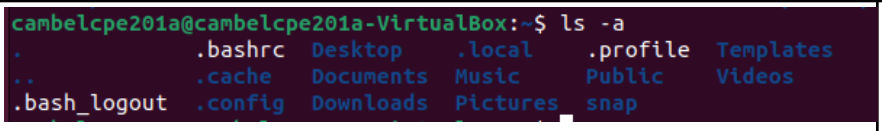
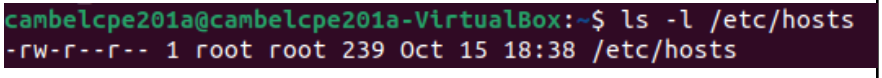
Command	Screenshot	Explanation

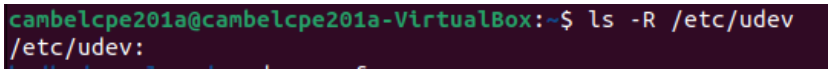
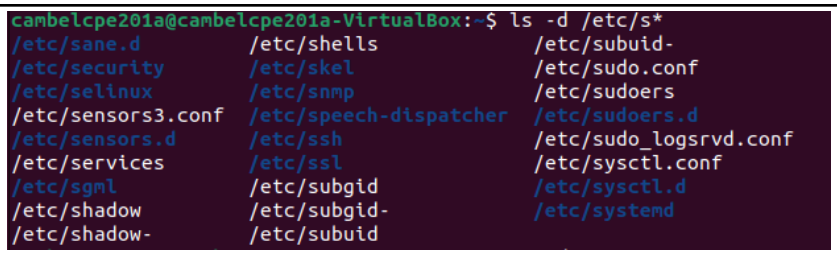
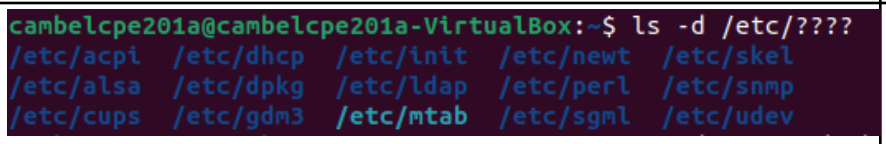
1. pwd		Displays the absolute path of the current working directory (/home/cambelcpe201a).
2. cd /		Changes the current working directory to the root directory (/).
3. pwd		Confirms that the current working directory is now the root directory (/)
4. cd		Navigates back to the logged-in user's home directory.
5. pwd		Verifies the user is back in the home directory (/home/cambelcpe201a).
6. cd /home		Changes directory

		to the system /home directory containing user profiles.
7. pwd		Confirms current path is /home.
8. cd ~		Uses the tilde shortcut (~) to change back to the user's home directory.
9. pwd		Prints current working directory path (/home/cambelcpe201a).
10. echo ~ ~sysadmin ~root ~mail ~nobody where sysadmin is your username		Expands tilde shortcuts to display home paths for specific system accounts /users.
11. cd ~root		Attempts to enter root home directory; denied due to

		lack of administrative permissions.
12. cd /usr/bin	`cambelcpe201a@cambelcpe201a-VirtualBox:~$ cd /usr/bin`	Changes working directory to /usr/bin where system executable binaries reside.
13. pwd	`cambelcpe201a@cambelcpe201a-VirtualBox:/usr/bin$ pwd` `/usr/bin`	Confirms current directory is /usr/bin.
14. cd /usr	`cambelcpe201a@cambelcpe201a-VirtualBox:/usr/bin$ cd /usr`	Navigates one level up to the /usr directory.
15. type ls	`cambelcpe201a@cambelcpe201a-VirtualBox:/usr$ type ls` `ls is aliased to 'ls --color=auto'`	Shows information about the ls command, indicating it is an alias for 'ls --color=auto'.
16. pwd	`cambelcpe201a@cambelcpe201a-VirtualBox:/usr$ pwd` `/usr`	Confirms current path is /usr.
17. cd /usr/share/doc	`cambelcpe201a@cambelcpe201a-VirtualBox:/usr$ cd /usr/share/doc`	Navigates directly to

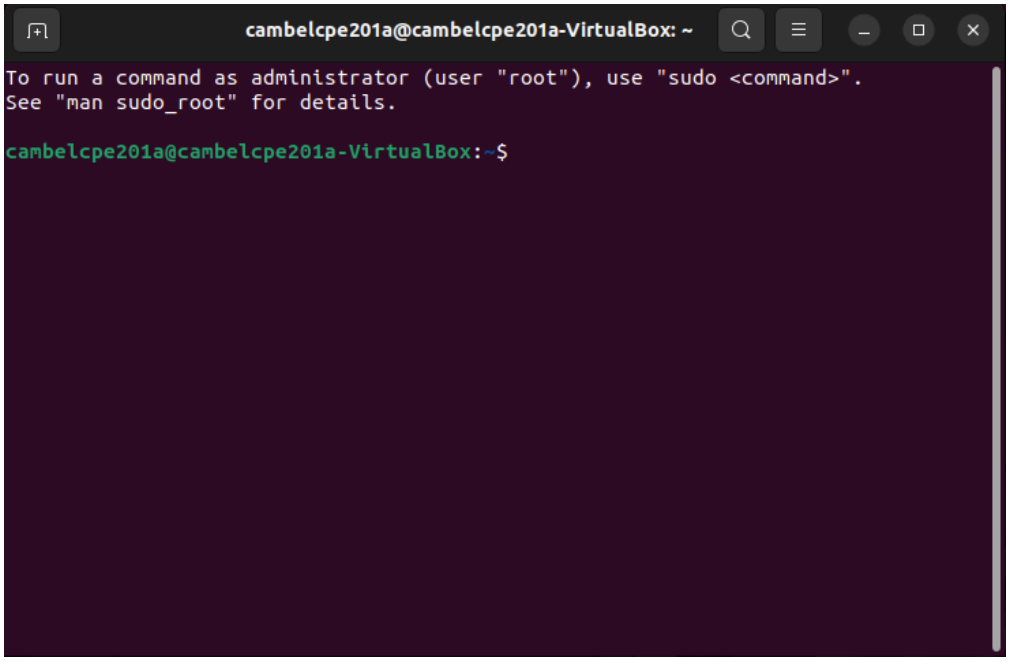
		/usr/share/doc using an absolute path.
18. pwd	cambelcpe201a@cambelcpe201a-VirtualBox:/usr/share/doc$ pwd /usr/share/doc	Verifies current directory is /usr/share/doc.
19. cd bash	cambelcpe201a@cambelcpe201a-VirtualBox:/usr/share/doc$ cd bas	Changes into the bash directory located inside /usr/share/doc using a relative path.
20. pwd	cambelcpe201a@cambelcpe201a-VirtualBox:/usr/share/doc/bash$ p /usr/share/doc/bash	Confirms current location is /usr/share/doc/bash.
21. cd ..	cambelcpe201a@cambelcpe201a-VirtualBox:/usr/share/doc/bash$	Uses parent relative path (..) to move back up to /usr/share/doc.
22. pwd	cambelcpe201a@cambelcpe201a-VirtualBox:/usr/share/doc$ pwd /usr/share/doc	Confirms location is /usr/share/doc.
23. cd ../dict	cambelcpe201a@cambelcpe201a-VirtualBox:/usr/share/doc$ cd ../	Navigates to the dict

		directory in the parent directory using relative path (../dict).
24. pwd		Verifies location is /usr/share/dict.
25. cd		Navigates back to the user's home directory.
26. ls		Lists all visible files and directories in the current folder.
27. ls -a		Lists all files and directories including hidden entries (those starting with a dot).
28. ls -l /etc/hosts		Displays detailed file permissions, owner, size, and date modified

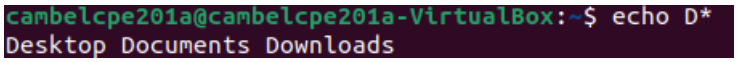
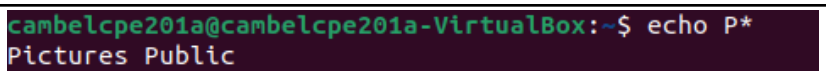
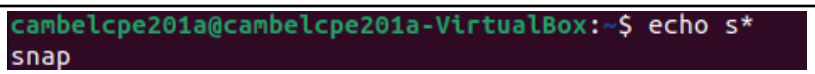
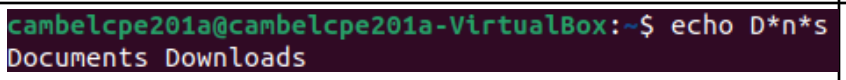
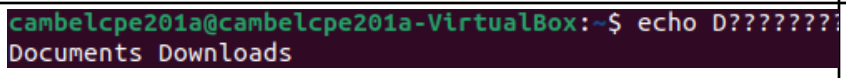
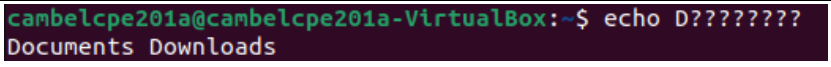
		for /etc/hosts.
29. ls -R /etc/udev		Recursively lists all files and contents inside /etc/udev directory.
30. ls -d /etc/s*		Lists directory names starting with 's' in /etc using globbing wildcard * without listing contents.
31. ls -d /etc/????		Lists directory names in /etc that are exactly 4 characters long using ? globbing.

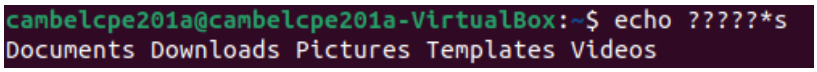
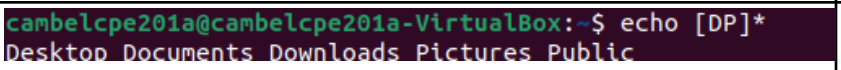
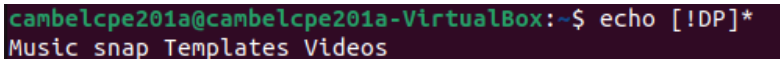
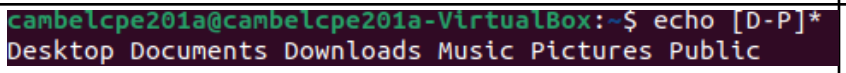
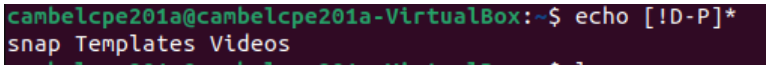
32. ls –d /etc/[abcd]*	cambelcpe201a@cambelcpe201a-VirtualBox:~$ ls -d /etc/[abcd]* /etc/acpi /etc/adduser.conf /etc/alsa /etc/alternatives /etc/anacrontab /etc/apg.conf /etc/apm /etc/apparmor /etc/apparmor.d /etc/apport /etc/appstream.conf /etc/apt /etc/avahi /etc/bash.bashrc /etc/bash_completion /etc/bash_completion.d /etc/bindresvport.blacklist /etc/binfmt.d /etc/bluetooth /etc/brlapi.key /etc/brltty /etc/brltty.conf /etc/ca-certificates /etc/ca-certificates.conf /etc/ca-certificates.conf.dpkg-old /etc/chatscripts /etc/console-setup /etc/cracklib /etc/cron.d /etc/cron.daily /etc/cron.hourly /etc/cron.monthly /etc/crontab /etc/cron.weekly /etc/cups /etc/cupshelpers /etc/dbus-1 /etc/dconf /etc/debconf.conf /etc/debian_version /etc/default /etc/deluser.conf /etc/depmod.d /etc/dhcp /etc/dictionaries-common /etc/dpkg	Lists directory names in /etc starting with letters a, b, c, or d using character set globbing.

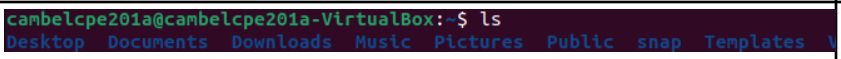
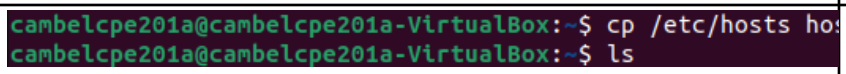
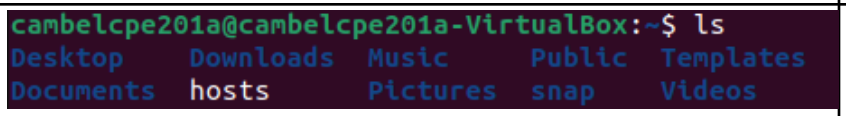
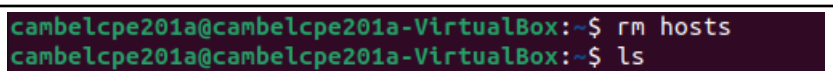
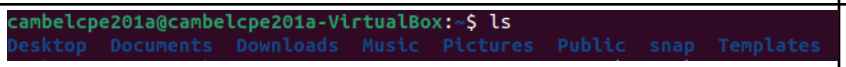
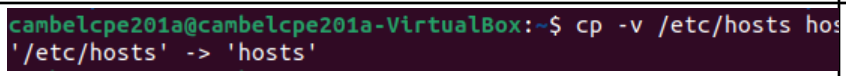
4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

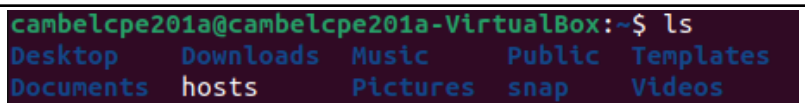
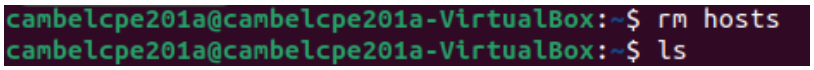
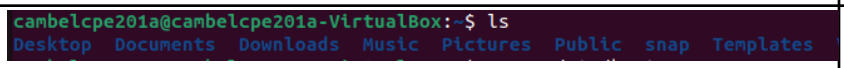
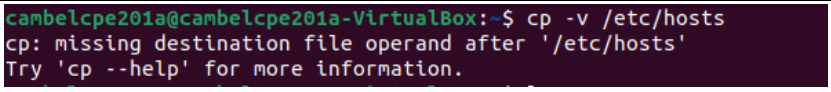
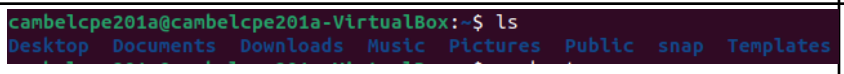
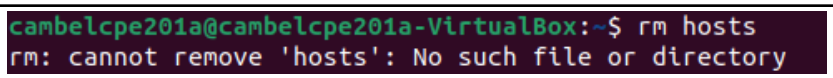


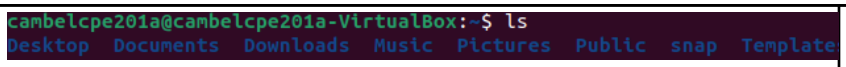
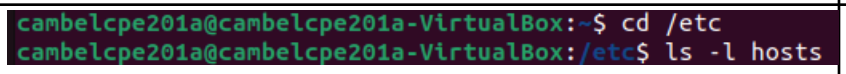
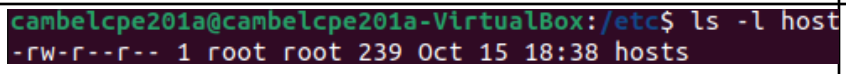
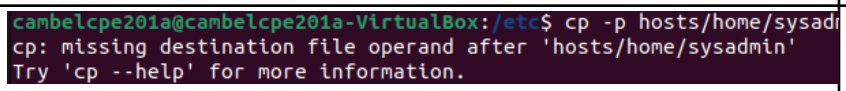
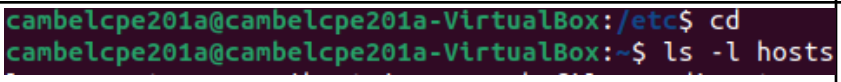
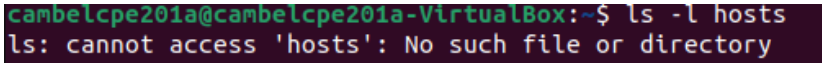
Command	Screenshot	Explanation
1. echo *	cambelcpe201a@cambelcpe201a-VirtualBox:~$ echo * Desktop Documents Downloads Music Pictures Public snap Templates Videos	Prints all filename

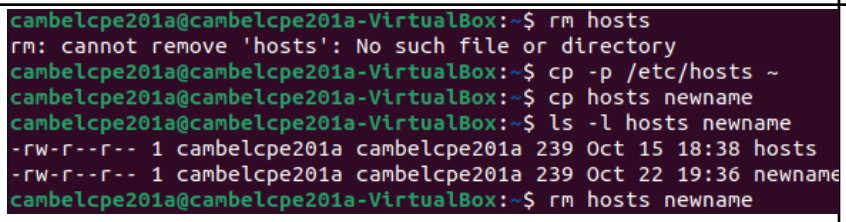
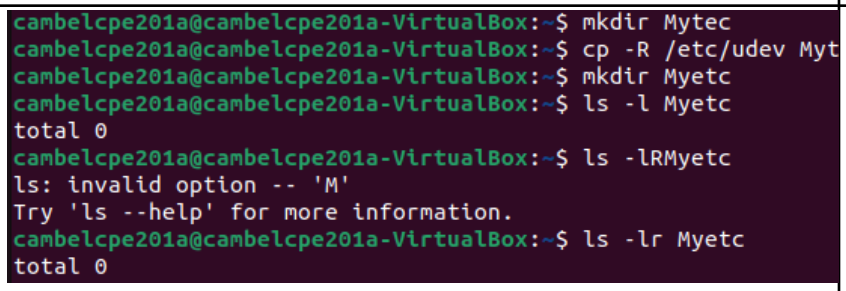
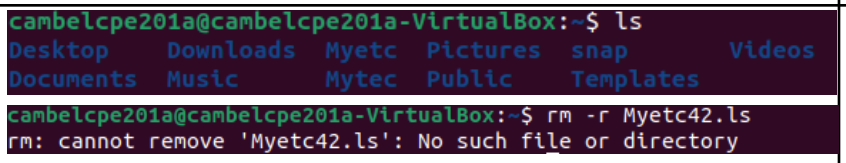
		s in the current directory using * wildcard extension.
2. echo D*		Prints filenames starting with upper-case letter 'D'.
3. echo P*		Prints filenames starting with upper-case letter 'P'.
4. echo *s		Prints filenames ending with letter 's'.
5. echo D*n*s		Prints filenames starting with 'D', containing 'n', and ending with 's'.
6. echo ??????		Prints filenames that are exactly 6 characters long using ? wildcard characters.
7. echo D????????		Prints filename

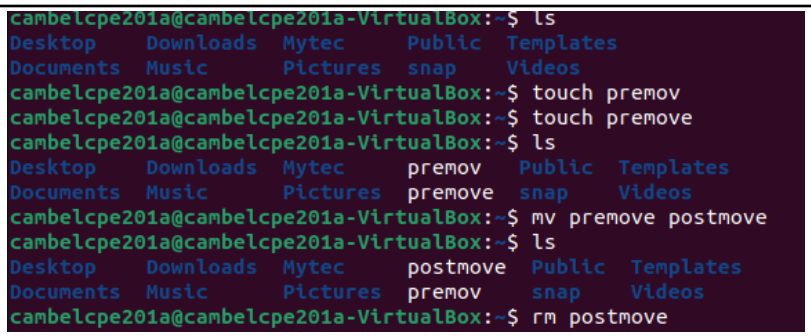
		s starting with 'D' followed by exactly 8 characters.
8. echo ?????*s		Prints filenames having at least 5 initial characters and ending with 's'.
9. echo [DP]*		Prints filenames starting with either 'D' or 'P'.
10. echo [!DP]*		Prints filenames NOT starting with 'D' or 'P' using negation operator (!).
11. echo [D-P]*		Prints filenames starting with letters in the alphabetical range D through P.
12. echo [!D-P]*		Prints filenames starting

		with letters outside the range D through P.
13. ls		Lists contents of current working directory.
14. cp /etc/hosts hosts		Copies /etc/hosts to local directory named 'hosts'.
15. ls		Verifies that file 'hosts' was created in current directory.
16. rm hosts		Removes the local file 'hosts'.
17. ls		Confirms deletion of local 'hosts' file.
18. cp –v /etc/hosts hosts		Copies file in verbose mode (-v) showing the source and destinati

		on mapping.
19. ls		Lists contents to verify file copying.
20. rm hosts		Deletes the copied local 'hosts' file.
21. ls		Confirms file removal.
22. cp –v /etc/hosts .		Attempts copy in verbose mode without specifying source/destination properly, causing error message.
23. ls		Lists directory contents.
24. rm hosts		Attempts to delete non-existent 'hosts' file, resulting in 'No such file or directory' error.

25. ls		Lists current directory contents.
26. cd /etc		Changes working directory to /etc.
27. ls -l hosts		Displays long listing format for /etc/hosts.
28. cp –p hosts/home/sysadmin		Fails due to missing space delimiter between source file and target directory path in syntax.
29. cd		Changes back to home directory and attempts to list local 'hosts' file.
30. ls –l hosts		Displays error since file 'hosts' was not copied previously due to

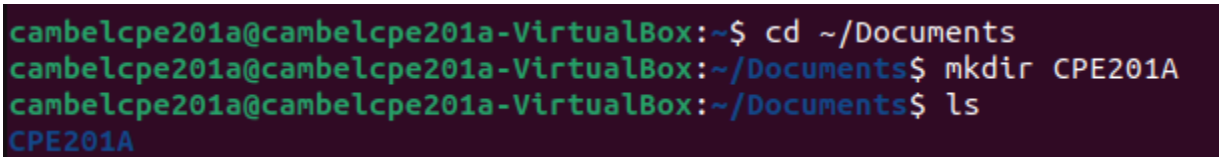
		syntax error.
31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname 34. ls –l hosts newname 35. rm hosts newname		Demonstrates preserving file properties (-p) when copying /etc/hosts to home (~), duplicating it as newname, listing details, and removing both.
36. mkdir Myetc 37. cp –R /etc/udev Myetc 38. ls –l Myetc 39. ls –lR Myetc		Creates Myetc directory, attempts recursive copying of /etc/udev, and tests long listing and recursive listing commands (-lR).
40. ls 41. rm -r Myetc42. ls		Shows directory contents with ls, attempts syntax-errored

		removal of Myetc, and verifies contents.
43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		Creates empty file with touch, lists contents, renames file using mv from premove to postmove, verifies change, and deletes file using rm.

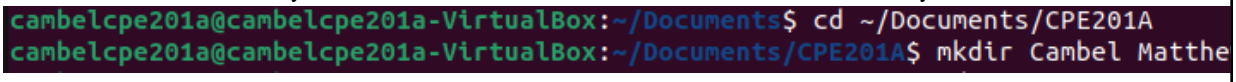
6. Supplementary Activity:

Copy screen shot(s) of the following tasks:

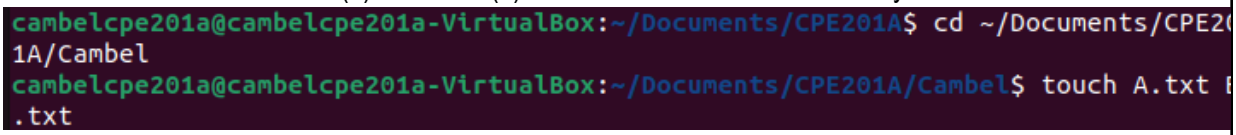
1. Create a directory CPE231 in the Documents directory.



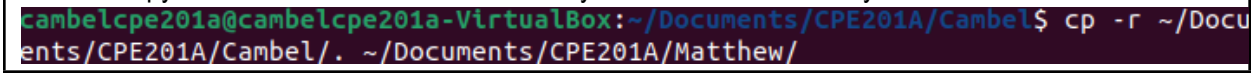
2. Create a directory <lastname> and <firstname> under CPE231 directory.



3. Create two text files: (1) A.txt and (2) B.txt in the <lastname> directory.



4. Copy the content of <lastname> directory to <firstname> directory.



5. Delete the content of <lastname> directory.

```
cambelcpe201a@cambelcpe201a-VirtualBox:~/Documents/CPE201A/Cambel$ rm -r ~/Docu
ents/CPE201A/Cambel/*
```

6. Create a directory CPE231_backup in the Documents directory.

```
cambelcpe201a@cambelcpe201a-VirtualBox:~/Documents/CPE201A/Cambel$ mkdir ~/Docu
ents/CPE201A_backup
```

7. Copy the content of CPE231 to CPE231_backup directory.

```
cambelcpe201a@cambelcpe201a-VirtualBox:~/Documents/CPE201A/Cambel$ cp -r ~/Docu
ents/CPE201A/. ~/Documents/CPE201A_backup/
```

8. Remove the CPE231 directory.

```
cambelcpe201a@cambelcpe201a-VirtualBox:~/Documents/CPE201A/Cambel$ rm -r ~/Docu
ents/CPE201A
```

7. Conclusion:

Through this activity, I learned how to navigate and manage the Linux file system using various terminal commands. I demonstrated navigation using absolute and relative paths with commands like pwd and cd, as well as special path representations such as the home directory (~), root (/), current directory (.), and parent directory (..). Furthermore, I practiced listing files with options like -a, -l, -R, and -d, and utilized wildcards and globbing characters (*, ?, []) to efficiently target specific files or directories. Lastly, I successfully demonstrated file and directory manipulation operations, including creating (mkdir, touch), copying (cp), moving/renaming (mv), and deleting (rm, rm -r) files and directories within Linux environments.

8. Assessment (Rubric for Laboratory Performance):